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Comparison of early versus delayed strategies for repair of congenital diaphragmatic hernia on extracorporeal membrane oxygenation☆

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ABSTRACT

Purpose: For the last seven years, our institution has repaired infants with CDH that require ECMO early after cannulation. Prior to that, we attempted to decannulate before repair, but repaired on ECMO if we were unable to wean after two weeks. This study compares those strategies.
Methods: From 2002 to 2016, 65 infants with CDH required ECMO. 67.7% were repaired on ECMO, and 27.7% were repaired after decannulation. Data were compared between patients repaired ≤5 days after cannulation ("early protocol", n = 30) and >5 days after cannulation or after de-cannulation ("late protocol", n = 35). We used Cox regression to assess differences in outcomes between groups.
Results: Survival for the early and late protocol groups was 43.3% and 68.8%, respectively ($p = 0.0485$). For patients that were successfully decannulated before repair, survival was 94.4%. Moreover, the early repair protocol was associated with prolongation of ECMO (16.8 ± 7.4 vs. 12.6 ± 6.8 days, $p = 0.0216$).
After multivariate regression, the early repair protocol was an independent predictor of both mortality (HR = 3.48, 95% CI = 1.28–9.45, $p = 0.015$) and days on ECMO (IRR = 1.39, 95% CI = 1.07–1.79, $p = 0.012$). All bleeding occurred in patients repaired on ECMO (29.5%, 13/44).
Conclusions: Our data suggest that protocolized CDH repair early after ECMO cannulation may be associated with increased mortality and prolongation of ECMO. However, early repair is not necessarily harmful for those patients who would otherwise be unable to wean from ECMO before repair. Further work is needed to better move towards individualized patient care.
Type of study: Treatment Study.
Level of evidence: Level III.

Despite significant advances in treatment strategies, between 20% and 40% of neonates with congenital diaphragmatic hernia (CDH) and persistent pulmonary hypertension will fail medical stabilization and require support with extracorporeal membrane oxygenation (ECMO) [1]. ECMO is often necessary to stabilize the patient prior to correction of the defect. While use of ECMO has been associated with increased survival in a subset of patients with severe disease [2,3], the optimal timing of repair once an infant has been cannulated is controversial.

Repair on ECMO removes the compressive effects of herniated abdominal viscera, in theory allowing postoperative lung recruitment and hopefully reducing pulmonary hypertension, and it provides cardiopulmonary support during the repair and in case of an exacerbation of pulmonary hypertension postoperatively [1,4,5]. For these reasons, proponents of early repair on ECMO suggest that repair may ultimately facilitate weaning from ECMO and that repair will be technically easier prior to development of anasarca and enlargement of the liver and spleen, which often occurs with prolonged ECMO owing to resuscitation and sequestration of blood products [1,6]. On the other hand, repair on ECMO is associated with a significantly increased risk of bleeding [7], and so repair after decannulation may lead to fewer surgical complications.

Our approach to repair of CDH in infants requiring ECMO has evolved. From 1992 to 2009 we favored a delayed repair protocol. Our strategy from the start of the study period in 2002 to 2009 was to attempt to decannulate before repair but to repair on ECMO if we were unable to wean after two weeks. However, for the last seven years,

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our institution has elected to repair the defects in these patients early after initiation of ECMO without first attempting to wean. This study was designed to compare those two strategies.

1. Methods

1.1. Patient population and study design

From March 2002 to May 2016, 165 infants with CDH were treated at the University of Michigan C.S. Mott Children's Hospital. ECMO support was required in 41.8% (69/165). Four of these patients were excluded: two for missing data regarding timing of either ECMO or operative repair and two because they underwent ECMO following repair, meaning their clinical course did not require a decision on whether to repair on ECMO or after decannulation. Only five patients in this study were transferred in after birth from another institution. One transferred on day of life #10 after first being cannulated on VA ECMO at the referring institution, and one transferred on day of life #12 after a successful VA ECMO run and decannulation at another hospital. The remaining three transferred on day of life #3. All of these patients underwent repair at the University of Michigan, and all five patients survived. Three of them were included in the late repair protocol group and two were included in the early repair protocol group.

Of patients requiring ECMO, 67.7% (44/65) underwent CDH repair while on ECMO. Of these, 65.9% (29/44) of infants were repaired ≤5 days after cannulation ("early repair protocol"), and the remaining 34.1% (15/44) were repaired >5 days after cannulation. An additional 27.7% (18/65) of patients were repaired following decannulation. For the purposes of our study, the "late repair protocol" group combined those patients repaired on ECMO >5 days after cannulation with the patients repaired following decannulation. The decision to combine these two groups was made to reflect the clinical realities of our protocol from 2002 to 2009: specifically, attempts to repair after decannulation yielded a percentage of patients that were unable to be decannulated and ended up being repaired "late" on ECMO. For the same reason, patients that died on ECMO before repair were included in the analysis to reflect the potential hazard of missed opportunity for repair with delayed surgery. It is important to note that the "late" repair protocol patients were all operated on between 2002 and 2009 and the "early" repair protocol patients were all operated on between 2009 and 2016. The two groups are reflective of management shifts within our institution, as described below.

A retrospective analysis was performed, and the "early" and "late" protocol groups were compared under institutional IRB approval (HUM00121205). Data were collected from the Vermont Oxford Network (VON) database and supplemented with specific prenatal, operative and pharmacologic data obtained through review of the electronic medical record. Our primary outcome was in-hospital mortality. Secondary outcomes were days on ECMO, bleeding complications, intracranial hemorrhage, hospital length of stay and days on the ventilator.

1.2. ECMO initiation and management

Following delivery, all infants underwent full, goal-directed resuscitation with a gentle ventilation strategy. Acceptable parameters were preductal O_2 saturations >85%, postductal PaO_2 > 30 mmHg, pH > 7.25, $PaCO_2$ = 45–65 mmHg and mean arterial pressure 40–45 mmHg. Acute deterioration with failure to meet these goals with conventional ventilator peak inspiratory pressures (PIP) 26–30 and mean airway pressures (MAP) >12 prompted consideration of high frequency jet ventilation (HFJV) or high frequency oscillatory ventilation (HFOV). PIP > 30 on the conventional ventilator or MAP > 22 on the oscillator with maximal inotropic/pressor support prompted initiation of ECMO. Infants at high-risk for severe pulmonary hypoplasia and pulmonary hypertension were identified prenatally based on liver up and prenatal lung-to-head ratio (LHR) ≤ 1.0, observed to expected (O/E) LHR ≤ 25% and/or MRI derived O/E total lung volume (TLV) ≤ 25%. ECMO circuits were on standby at delivery for these infants. Since ECMO utilization in this population has been near 100% in our experience, we elected to initiate ECMO immediately after the initial resuscitation to avoid barotrauma and provide a smooth early transition to ECMO when the infants were relatively stable after delivery. However, infants on the most severe end of the spectrum underwent our trial of life or Severe Pulmonary Hypoplasia and Evaluation for Resuscitative Efforts (SPHERE) protocol in an effort to identify those that have pulmonary hypoplasia that is incompatible with life and not expected to improve with time on ECMO. If the infant was unable to obtain a pH >7.0, $PaCO_2$ ≤ 100, preductal SaO_2 ≥ 80%, and a PaO_2 ≥ 40 on ventilator support utilizing the maximum settings (described above) with appropriate sedation and optimization of blood pressure over the first 2 h of life, or if the infant was clearly moribund or coding, redirection of goals of care to comfort was strongly considered. If, however, the baby met these criteria at any point during the two hours of observation, ECMO cannulation was performed. A small number of patients were born using an ex-utero intrapartum treatment (EXIT) to ECMO protocol, but despite good outcomes that approach was abandoned owing to unclear benefits and potential maternal risks. For all patients, steroids were utilized to empirically treat adrenal insufficiency, and inhaled nitric oxide (iNO) was utilized sparingly as a bridge to ECMO support. ECMO was not offered to infants with serious coexistent genetic anomalies, uncorrectable cyanotic heart disease, moribund condition, severe intracranial hemorrhage, or weight < 1.5 kg.

From 2002 to 2009, our protocol was to attempt to wean ECMO before repair but to repair CDH on ECMO if the patient was unable to wean after about 2 weeks of support. Two weeks was selected because in our experience patients that have required the circuit for that long have a diminishing likelihood of successfully weaning from ECMO, and the hope was that repair would provide a boost in lung recruitment that was enough to help those patients come off ECMO. If decannulation was successful, repair was further delayed until respiratory insufficiency and hemodynamic status improved and plateaued without expectations for further improvement. In particular, we attempted to stabilize infants on conventional ventilators prior to repair so that we had means for escalation (other than re-cannulation of ECMO) if they became sicker for a period after surgery. Since 2009, the protocol for CDH repair for infants requiring ECMO is to repair them as early as 24–48 h following cannulation.

Patients were maintained on ECMO with ACT goals of 210–230. Six hours prior to surgery, Amicar was bolused at 100 mg/kg, and it was continued as an infusion at 33 mg/kg/h for 24 h. Furthermore, ACT goals were adjusted to 170–190, platelets were maintained >100,000 and hematocrit was kept >40 before repair on ECMO. The lowered ACT goal was continued for the duration of the amicar infusion. Repairs on ECMO were performed through subcostal incisions with meticulous hemostasis, and abdominal wall silos were routinely placed unless the abdominal domain was capacious [8,9]. Of note, head ultrasounds were performed every Monday, Wednesday and Friday to evaluate for intracranial hemorrhage. Echocardiograms were obtained on day 1 to exclude congenital heart disease, at 1 month, and, otherwise, as needed to assess pulmonary hypertension. Diuresis with lasix was typically initiated after 48–72 h. We first assessed for trial off ECMO following optimization of hemodynamics, volume status, and pulmonary mechanics, as appropriate, after approximately one week of support, and there were no differences in attempts to decannulate between protocol groups.

1.3. Statistical analysis

Data are described as mean $\pm$ standard deviation or percentage. Patient characteristics were compared by timing of repair. Continuous variables were compared using the Mann–Whitney test, and categorical variables were compared using Chi Square or Fisher's exact test as appropriate.

Cox regression analysis (proportional hazards analysis) was used to assess the relationship between the two different timing of repair protocols and survival. Time in this study was defined as years of life at death. Kaplan–Meier survival curves were initially examined to explore the appropriateness of the Cox regression (Fig. 1). Since the curves appeared to deflect, Cox regression as a next step was deemed appropriate to assess survival. Next, the association of time with each potential clinically relevant covariate was examined by separate Kaplan–Meier survivor functions (if categorical) or single factor Cox regression (if continuous). Variables considered included the following: estimated gestational age, birth weight, APGAR scores at 5 min, defect type (A–D), left-sided defect, liver up, prenatal diagnosis, pressor/inotrope support at the time of repair or death, pulmonary vasodilator use at the time of repair or death and major cardiac defects [10]. Covariates which were significantly associated at a p-value <=0.30 were entered into the initial multivariable Cox model. Next, in a stepwise fashion, covariates which were not associated with time at the 0.50 level were removed via a backward likelihood ratio procedure, which tests whether or not the "smaller" nested model is better. Results from Cox regression analyses are reported as hazard ratios (HRs). Since duration of ECMO therapy is a nonnormally distributed count variable, factors associated with duration of ECMO therapy were assessed via multivariable negative binomial regression. Results are reported as incidence risk ratios (IRRs).

All tests were two-sided with the level of significance set at 0.05. All analyses were performed using STATA13 (StataCorp, College Station, TX).

2. Results

Table 1 compares clinical and operative characteristics between the early and late repair protocol groups. The early repair protocol group was comprised of more defects with liver up (83.3% vs. 57.1%, $p = 0.0314$), possibly owing to a trend towards more right sided defects (30.0% vs. 14.3%, $p = 0.1432$). There were no differences between groups in terms of birth weight, gestational age, APGAR scores, severe cardiac defects, prenatal diagnosis or defect size. Support method was VA ECMO for all patients with the exception of 2 patients in the late repair protocol group who underwent VV ECMO.

Infants in the early repair protocol group were cannulated for ECMO at an average of 0.53 ± 0.90 days of life (range 0–4, median/mode = 0 days of life). They were repaired on average by day of life two and spent 16.8 ± 7.4 days on ECMO (Tables 1 and 2). One patient in the early protocol group died before repair, and 24.1% of patients (7/29) were unable to wean from ECMO following repair. The patient who died before repair was decannulated after three days on ECMO owing to improving gas exchange and a run complicated by severe hemolysis with subsequent renal tubular injury. While initially relatively stable off ECMO, the patient deteriorated over the next several hours owing to severe pulmonary hypertension. Owing to a poor prognosis from his concomitant cardiac disease, which was more severe than initially recognized, the patient was transitioned to comfort care. Of the patients unable to wean from ECMO after repair, all seven had severe pulmonary hypertension, two had a poor prognosis from additional multisystem organ failure (MSOF) and one had severe cardiac disease which couldn't

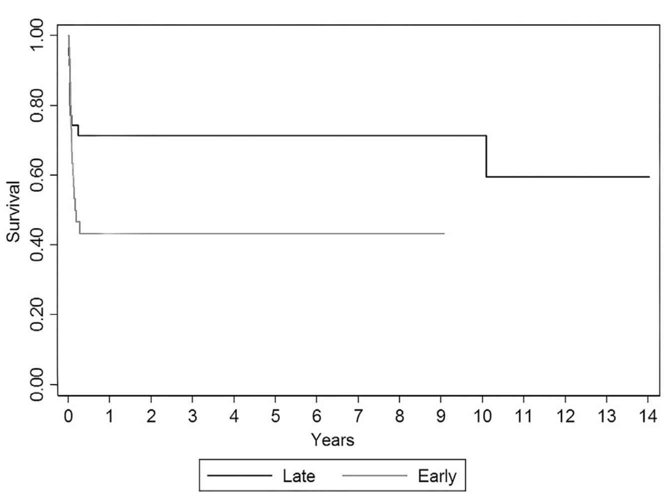

Fig. 1. Kaplan–Meier survival curves by timing of repair. Log-rank test: $\chi^2 = 3.70$, p = 0.054.

Table 1
Clinical and operative characteristics.

Variable	Early Repair Protocol (n = 30)	Late Repair Protocol (n = 35)	*P*-value
Birth Weight (g)	2902.1 ± 558.4	3071.0 ± 493.3	0.2047
Gestational Age (wk)	38.0 ± 2.0	38.4 ± 1.6	0.3762
APGARS 1 min	3.2 ± 1.8	3.1 ± 2.1	0.9100
APGARS 5 min	5.8 ± 1.9	5.2 ± 2.0	0.1973
pH on Initial ABG	7.21 ± 0.12	7.23 ± 0.13	0.5337
Severe Cardiac Defect	20.0% (6/30)	8.6% (3/35)	0.2821
Prenatal Diagnosis	90.0% (27/30)	82.4% (28/34)	0.4831
Polyhydramnios	24.1% (7/29)	21.9% (7/32)	1.0000
CDH Defect Type			
A	3.7% (1/27)	6.1% (2/33)	1.0000
B	33.3% (9/27)	18.2% (6/33)	0.2348
C	44.4% (12/27)	42.4% (14/33)	1.0000
D	18.5% (5/27)	33.3% (11/33)	0.2481
Left Sided Defect	70.0% (21/30)	85.7% (30/35)	0.1432
Liver Up	83.3% (25/30)	57.1% (20/35)	0.0314
Repair Day of Life	2 ± 1.5	16.4 ± 6.0	<0.0001
Thoracoscopic Repair	0.0% (0/29)	6.1% (2/33)	0.4939
Patch Repair	71.4% (20/28)	84.8% (27/33)	0.3746
VA ECMO	100% (30/30)	94.3% (33/35)	0.4952
Repair on ECMO	96.7% (29/30)	42.9% (15/35)	<0.0001
Repair After ECMO	0.0% (0/30)	51.4% (18/35)	<0.0001
Ventilator Mode at Time of Repair			
PC/SIMV	11.5% (3/26)	3.4% (1/29)	0.3346
PC/AC	88.5% (23/26)	86.2% (25/29)	1.0000
CPAP	0.0% (0/26)	10.3% (3/29)	0.2384
HFOV/Jet	0.0% (0/26)	0.0% (0/29)	n/a
Use of Flolan[a]	50.0% (15/30)	22.9% (8/35)	0.0366
Use of iNO[a]	66.7% (20/30)	40.0% (14/35)	0.0465
Pulmonary Vasodilator at Time of Repair or Death (if not repaired)	3.6% (1/28)	20.6% (6/34)	0.1160
Pressor/Inotrope at Time of CDH Repair or Death (if not repaired)	63.3% (19/30)	29.4% (10/34)	0.0114
Dopamine Use[a]	100% (30/30)	94.3% (33/35)	0.4952
Epinephrine Use[a]	46.7% (14/30)	28.6% (10/35)	0.1973
Hydrocortisone Use[a]	73.3% (22/30)	57.1% (20/35)	0.2020

[a] Anytime during initial hospitalization.

Table 2
Outcomes.

Variable	Early Repair Protocol (n = 30)	Late Repair Protocol (n = 35)	*P*-value
Mortality	56.7% (17/30)	31.4% (11/35)	0.0485
Days on ECMO	16.8 ± 7.4	12.6 ± 6.8	**0.0216**
Died Before Repair	3.3% (1/30)	5.7% (2/35)	1.0000
Unable to Wean from ECMO	26.7% (8/30)	17.1% (6/35)	0.3815
Number of ECMO Circuit Changes	30.0% (9/30)	2.9% (1/35)	0.0040
Takeback for Bleeding	30.0% (9/30)	8.6% (3/35)	0.0513
Bleeding Not Requiring Takeback	0.0% (0/30)	2.9% (1/35)	1.0000
LOS	62.8 ± 58.6	68.5 ± 48.8	0.6771
Days on the ventilator	48.2 ± 47.7	46.1 ± 44.4	0.8551
Bacterial Infection	16.7% (5/30)	28.6% (10/35)	0.3770
ICH	17.2% (5/29)	11.4% (4/35)	0.7200

be repaired in the setting of such severe pulmonary hypertension. In all cases, comfort care was initiated after discussion with the family. Importantly, a higher percentage of patients in the early group required ECMO circuit changes (30.0% vs. 2.9%, $p = 0.0040$), possibly reflecting the increased time on ECMO.

In contrast, infants in the late protocol group were cannulated for ECMO at an average of 0.66 ± 1.26 days of life (range 0–7, median/mode = 0 days of life). They were repaired on day of life 16.4 ± 6.0 and spent 12.6 ± 6.8 days on ECMO. Two patients in the late protocol group died before repair. Care was withdrawn on one secondary to severe MSOF, and the second developed a grade III intraventricular hemorrhage on ECMO with hydrocephalus and severe pulmonary hypertension. Four patients (12.1%) were unable to wean from ECMO after repair: two owing to severe MSOF and severe pulmonary hypertension and two for extensive NEC. Considering only patients that were repaired late on ECMO, the time on circuit was 17.9 ± 5.6 days. Patients repaired off ECMO spent an average of 7.6 ± 2.8 days on ECMO, and subsequently underwent surgery 8.4 ± 7.0 days after decannulation.

The proportion of patients on inotropes/pressors and pulmonary vasodilators at the time of repair are listed in Table 1. Briefly, the late repair protocol group used significantly fewer pulmonary vasodilators (Flolan, iNO) during their hospital course but trended towards being on them more frequently at the time of repair or death (3.6% vs. 20.6%, $p = 0.1160$). The fact that few patients in the early protocol group were on pulmonary vasodilators at the time of repair is reflective of the fact that we don't typically start iNO or flolan until around the time we are ready to trial off and patients demonstrate a need for it, and if any patients had been on iNO as a bridge to ECMO, it would have been discontinued once they were on the circuit. Likewise, significantly fewer patients in the late protocol group were on pressors and inotropes at that same time point (63.3% vs. 29.4%, $p = 0.0114$), reflecting either greater clinical stability at the time of repair or a shift in critical care management of these patients over time.

Bleeding complications occurred only in patients repaired on ECMO (29.5%, 13/44). Of those, 92.4% required operative intervention (Table 2). There was a trend towards a higher incidence of reoperation for bleeding in the early repair protocol group (30.0% vs. 8.6%, p = 0.0513), which likely reflects the fact that a higher proportion of patients were repaired on ECMO in this group. Reoperation consisted of abdominal exploration, often facilitated by the silo placed at the end of the original operation, in 66.7% of cases (8/12). A thoracotomy was required for exploration in the chest in 33.3% of cases (4/12). Two patients required two reoperations for bleeding: a thoracotomy on one occasion and an abdominal exploration on the other. Of the patients receiving reoperation for bleeding, 66.7% (8/12) of the patients died.

Survival was higher in the late repair protocol group (43.3% vs. 68.6%, $p = 0.0485$) (Table 2). A Kaplan–Meier survival curve is presented in Fig. 1. Patients repaired after ECMO had a 94.4% survival (17/18) compared with a 45.5% (20/44) survival for those repaired on ECMO ($p = 0.0004$). After multivariable Cox regression analysis accounting for covariates known to increase clinical risk of death in CDH, the early repair strategy was associated with increased risk of in-hospital mortality (Table 3, HR = 3.48, 95% CI = 1.28–9.45, $p = 0.015$; Fig. 2).

As mentioned above, the early repair protocol group spent significantly more days on ECMO when compared to the late repair protocol group. After multivariable regression for a number of clinical markers of severity, this association still held (Table 4, IRR = 1.39, 95% CI = 1.07–1.79 $p = 0.012$). After multivariate correction, timing of repair protocol did not affect risk of intracranial hemorrhage ($p = 0.305$), days on the ventilator ($p = 0.397$), or hospital LOS ($p = 0.899$).

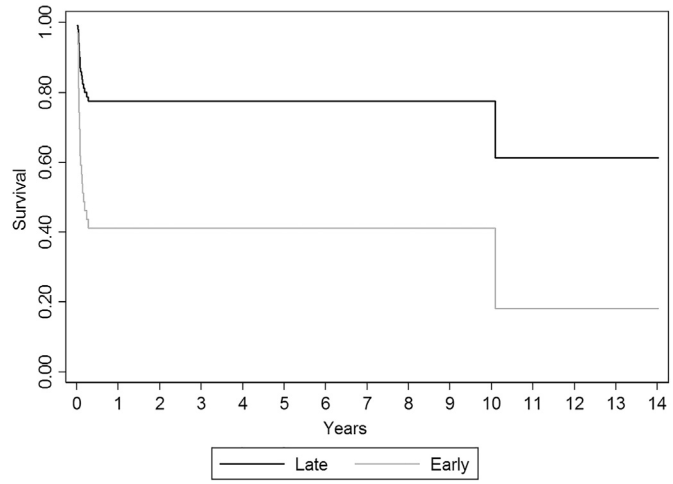

Fig. 2. Cox Proportional Hazard model for survival by timing of repair while controlling for other covariates.

When patients repaired after decannulation were excluded, there was no difference between survival or time on ECMO for those repaired early or late on ECMO: survival was 43.3% (13/30) for the early repair protocol group and 41.2% (7/17) for the late repair group ($p = 1.000$). Time on ECMO was 16.8 ± 7.4 days for the early protocol group and 17.9 ± 5.6 days for the late protocol group (p = 0.5846).

3. Discussion

Among infants with CDH, the patients at highest risk for mortality are those who require ECMO, where reported survival rates are between 45% and 65% [11,12]. There is a large volume of data to suggest that preoperative stabilization before repair is beneficial in CDH patients who do not require ECMO [12–17], but timing of repair once ECMO is initiated remains controversial. ECMO was first used at the University of Michigan in 1981, and we remain a large volume center with respect to use of ECMO in severe CDH. This study is unique in that it compares contemporary patient populations at a single institution where there was a paradigm shift in protocol that was followed by all surgeons in the group, thereby eliminating selection bias in terms of which patients were repaired early or late. Our data demonstrate a significantly increased risk of death and prolongation of ECMO when a protocol for early repair on ECMO is followed, even after controlling for factors associated with increased mortality. This seems to be driven by the fact that those who can be weaned from ECMO have significantly higher survival and decreased risk of bleeding when the operation is performed off ECMO. While some patients require repair on ECMO because of inability to wean from extracorporeal support, outcomes are similar regardless of when the surgery is performed, so long as the patient is on the circuit

Table 3
Cox regression analysis for time to death.

	HR	95% Confidence Interval	p-value
Early repair protocol	3.48	1.28–9.45	0.015
Pressor/inotrope support at time of repair or death	0.33	0.11–0.97	0.044
5 min APGAR	0.57	0.44–0.73	<0.0001
Severe Cardiac Defect	3.98	1.35–11.68	0.012

Table 4
Multivariable regression for duration of ECMO.

	IRR	95% Confidence Interval	p-value
Early repair protocol	1.39	1.07–1.79	0.012
Pressor/inotrope support at time of repair or death	0.81	0.63–1.04	0.091
5 min APGAR	0.94	0.89–0.99	0.034
Liver up	1.32	1.02–1.71	0.036
Severe Cardiac Defect	1.38	0.99–1.93	0.055

at the time of repair. In other words, indiscriminate early repair appears harmful to those for whom repair could be delayed until after decannulation, but early repair on ECMO is not necessarily harmful for patients that are otherwise unable to wean from ECMO before repair.

Early in our experience with ECMO at Michigan, a study was performed wherein the practice of immediate repair (from 1988 to 1992, n = 66) was compared to delayed repair (from late 1992 to early 1996, n = 33) [12]. While that study included all patients with CDH, 17 patients from the immediate repair group and 11 patients from the delayed repair group required ECMO prior to repair and so are analogous to our current analysis. Survival was higher for infants that underwent delayed repair (35.3% vs. 54.5%), and it reached 100% for those that were able to be weaned off of ECMO prior to repair. While these numbers are too small to show significant differences in the subset with ECMO before repair, similar findings have been reported at two Canadian hospitals [18,19], and the trends mirror what we are observing in our current population of patients.

Another single institution study by Partridge et al. shows survival of 43.9% (18/41) for repair on ECMO compared with 100% survival for repair after ECMO (17/17, $p < 0.0001$) [20]. Similarly, they demonstrated that bleeding occurred exclusively with repair on ECMO (29% vs 0%, $p < 0.01$) and that repair on ECMO prolonged duration of ECMO (18.8 vs. 13.0 days, $p = 0.02$). Unlike our study, however, there were no stated institutional protocols and no multivariate correction for mortality or length of time spent on ECMO, so there is likely a more prominent selection bias in that study. Nevertheless, the data are remarkably similar to our own.

Data from the Congenital Diaphragmatic Hernia Study Group (CDHSG) also support our conclusion that an attempt to decannulate before repair on ECMO is warranted [4]. The authors examined 636 patients that underwent repair and ECMO between 1995 and 2005. After multivariate cox regression to account for covariates associated with mortality, the authors demonstrated that repair on ECMO was associated with a hazard ratio of 1.41 (95% CI = 1.03–1.92, $p = 0.03$) compared to those that underwent repair after ECMO. The authors speculated that the mortality differences were related to more frequent bleeding complications with repair on ECMO, but their data did not allow the issue of bleeding to be directly addressed. While the CDHSG paper did not compare systematic protocols for repair, it did attempt to control for differences in clinical characteristics and it suggests that there may be a benefit to attempting decannulation before repair.

In contrast to these data, two studies in the literature favor early repair on ECMO [1,6]. The first by Dassinger et al. compared single institution data (n = 34) for patients repaired after an average of 2.3 days on ECMO with data from the CDHSG (where the average time to repair on ECMO was 6.9 days). The authors reported that their institutional survival was superior to that seen in the registry (71% vs. 49%, p = 0.16), and they concluded that early repair was associated with better survival. The second study compared patients repaired early on ECMO (n = 11), late on ECMO (n = 18), and postdecannulation (n = 17) [1]. They found that survival was 73%, 50% and 64% for these groups, respectively. Additionally, patients repaired within 72 h of cannulation were decannulated 6 days earlier (12 vs. 18 days, p = 0.009) and had significantly fewer circuit changes. Neither study, however, corrected for differences between groups, and it is unclear how much of the deviation from our results can be accounted for by differences in clinical management between institutions. Still, our study best reflects real world protocols and decision making since repair after ECMO is analyzed alongside late repair on ECMO.

The reasons for our findings are likely several-fold. First, all bleeding complications were observed with repair on ECMO. While our incidence of bleeding is similar to that from other contemporary reports and better than historical data before initiation of preoperative protocols utilizing antifibrinolytic therapy [1,6,7], we still observed 66.7% mortality in our 12 patients that required reoperation for bleeding. This highlights the need for improved strategies for anticoagulation management while on ECMO, and it suggests that some of the risk with early repair would be alleviated if better management strategies were developed. Second, data suggest that infants with CDH may often suffer from significantly decreased pulmonary compliance in the initial period following repair, which may contribute to worsened hypoxemia and acidosis [21]. While ECMO has been shown to increase compliance, the specific effects of repair on pulmonary function in infants on ECMO and the interrelationship between the two are uncertain. Moreover, when compliance does increase following surgery it increases to a much greater extent when patients have undergone a period of preoperative stabilization [22]. The rationale behind our protocol to repair patients on ECMO if they have been unable to decannulate by about 14 days is that the repair may provide the boost in pulmonary function that they need to wean off circuit. The data from Nakayama et al. suggest that the degree of improvement in respiratory compliance may be two-fold greater after an average 6 day period of stabilization [22], so it is possible that less of a benefit is provided if the repair is performed early while on ECMO.

The differences we observe in time on ECMO are most likely related to the fact that patients that are able to wean before repair spend less time on the circuit than those that cannot. Excluding patients that were repaired after decannulation, we see no differences between the early and late protocol groups in terms of either mortality or duration of ECMO. However, this is still clinically important. It suggests that surgeons who follow a protocol whereby all patients are repaired early manage to keep the subset of patients that could have weaned off if given a chance (an uncertain number but one that can be estimated at 51% based on the preceding 7 years of data) on ECMO longer. At the same time, perhaps by extending time on the circuit and exposure to all the associated risks therein, the survival benefits these patients would have otherwise seen by being repaired off circuit after a period of stabilization are eliminated. This loss of benefit for the healthier patients that could otherwise have weaned off ECMO does not seem to be negated by delaying repair in patients unable to wean from ECMO. Unfortunately, we do not always have a reliable way to determine which patients will be able to wean off ECMO and which will not [23,24]. That is an area that requires further investigation, as optimal timing for repair is likely patient-specific, and might even be before initiation of ECMO, as some have suggested [25].

This paper is unique in that rather than comparing outcomes for repair early versus late on ECMO with repair off ECMO, it compares the *strategy* of attempting to decannulate patients before repair with performing a planned early repair on ECMO, making the analysis applicable to clinical practice. Moreover, since these two divergent strategies have been universally applied by all surgeons at our institution during distinct time periods, this study eliminates many aspects of selection bias inherent in patient management. This is also one of the few studies examining the issue that controls for covariates associated with adverse outcomes through use of a multivariable model. Still, our data suggest that a one-size fits all protocol, while easier to implement, may not be ideal, and results may be improved with better evidence-based algorithms for individualized treatments, risk adjustment and circuit management strategies.

This paper has limitations inherent to a retrospective analysis examining a relatively rare disease. Despite confidence that we mitigated selection bias through an institutional shift in practice, this is not a randomized controlled trial, nor are the two groups equivalent. We make the assumption that the early repair protocol group included several patients over its seven-year implementation that were moderately severe and could have weaned off of ECMO if given the chance, but we have no way to confirm that with our data. Rather, we are only able to estimate the size of that group based on comparison to historical data leading up to the protocol change. Moreover, because the study period traversed 14 years, there is an unquantifiable change over time in ECMO management (including specifics of assessment for decannulation), equipment (including a transition to centrifugal pumps and more efficient oxygenators in the second era) and critical care. We are also, unfortunately, unable to include O/E LHR and TLV in our models because not all patients were diagnosed prenatally, and not all patients underwent preoperative risk stratification and counseling. Lastly, we did not assess the long-term outcomes in this cohort of patients.

4. Conclusions

In conclusion, this study suggests that an early repair on ECMO protocol is associated with more days on ECMO and increased in-hospital mortality. We should emphasize that our data show that early repair on ECMO is not necessarily harmful to patients that would not otherwise be able to wean from the circuit before repair; rather, it is the protocol of repairing everyone early on ECMO that seems to impact outcomes. A strategy whereby an attempt is made to decannulate prior to repair may benefit the subset of patients that is able to wean from ECMO and delay repair until they are clinically optimized. Further study is needed to reliably predict which patients will be able to wean before repair. Moving towards individualized patient care will require further work in risk adjustment and better data to inform surgical decision making. Our data also suggest that bleeding remains an important issue for patients repaired on ECMO, and new strategies are required to further reduce this complication and its consequences.

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